

DATA ANALYST INTERNSHIP

Task 1: Data Cleaning and Preprocessing — Summary Report

1. Overview

Intern	Data Analyst Intern
Organization	DataX Labs
Task	Task 1 — Data Cleaning and Preprocessing
Dataset	Sales Data (Customer Transactions)
Tools Used	Python 3.12, Pandas, NumPy
Date	29 June 2026
GitHub Repo	github.com/[your-username]/datax-labs-internship

2. Raw Dataset Profile

The raw dataset contained 203 rows and 9 columns representing customer sales transactions. The following data quality issues were identified before cleaning:

Issue	Column(s) Affected	Count
Missing values	Age, Sales Amount, Rating	20 + 11 + 21 = 52
Inconsistent text casing	Customer Name, Gender, Country	~60 rows
Non-standard gender values	Gender	M, F, male, female, etc.
Non-standard country values	Country	US, USA, United States, india
Mixed date formats	Purchase Date	dd/mm/yyyy, dd-mm-yyyy, yyyy-mm-dd
Incorrect data types	Age, Rating (float)	All rows
Inconsistent column names	All columns	Spaces, mixed case

3. Cleaning Steps Performed

Step 1 — Removed Duplicate Rows

Used `.drop_duplicates()` to remove exact duplicate rows. Ensured data integrity by keeping only unique records.

```
df.drop_duplicates(inplace=True)
```

Step 2 — Standardized Column Headers

Renamed all column headers to lowercase with underscores for consistency and ease of use in code.

```
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')
```

Step 3 — Standardized Text Columns

Applied `.str.title()` to Customer Name. Mapped Gender values (M/F/male/female) to Male/Female. Mapped Country variants (US/United States/india) to standardized values (USA/India).

```
df['gender'] = df['gender'].str.lower().map(gender_map)
```

Step 4 — Handled Missing Values

Age: filled with column median (robust to skew). Sales Amount: filled with per-product median (preserves category patterns). Rating: filled with mode (most frequent value).

```
df['age'] = df['age'].fillna(df['age'].median())
```

Step 5 — Fixed Data Types

Converted Age and Rating from float64 to int64. Ensured Sales Amount is float. Parsed all Purchase Date formats to a uniform dd-mm-yyyy string.

```
df['age'] = df['age'].astype(int)
```

Step 6 — Outlier Detection

Applied IQR method on Sales Amount. No outliers were found outside the expected range. Results were logged for documentation purposes.

```
Q1, Q3 = df['sales_amount'].quantile([0.25, 0.75])
```

4. Before vs. After Comparison

Metric	Before Cleaning	After Cleaning
Total Rows	203	203
Duplicate Rows	0	0
Missing Values (total)	52	0
Gender unique values	M, F, male, female, Male, Female, FEMALE	Male, Female
Country unique values	India, india, INDIA, USA, US, United States	India, USA
Age data type	float64	int64
Rating data type	float64	int64
Date format	Mixed (3 formats)	Unified dd-mm-yyyy
Column names	Mixed case with spaces	Lowercase with underscores

5. Key Learnings

- Real-world datasets almost always contain dirty data — nulls, inconsistencies, and type mismatches.
- Choosing the right fill strategy for missing values matters: median for continuous skewed data, mode for categorical/ordinal.
- Standardizing text fields early prevents groupby errors and incorrect analysis downstream.
- IQR-based outlier detection is a robust, non-parametric way to spot anomalies without assuming normality.
- pandas Copy-on-Write in v2.x requires assignment-style updates; inplace on chained references is unreliable.

6. Deliverables

File	Description
raw_sales_data.csv	Original dataset with intentional data quality issues
cleaned_sales_data.csv	Final cleaned and preprocessed dataset
data_cleaning.py	Python script with all cleaning steps and inline comments

changes_summary.txt	Plain-text log of all changes made
Task1_DataCleaning_Report.pdf	This summary report
README.md	Project overview and instructions to reproduce

Submitted as part of the DataX Labs Data Analyst Internship — Task 1